

Overview: Pruritus — A Comprehensive Clinical Guide

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- Overview: Pruritus — A Comprehensive Clinical Guide
- 1 Chapters • Premium Tables • Key Teaching Points
 - OSCE Pearls • MCQ Traps • High-Yield Facts

2025 Edition

TABLE OF CONTENTS

Chapter 1 Overview: Pruritus — A Comprehensive Clinical Guide

Overview: Pruritus — A Comprehensive Clinical Guide

Pruritus is defined as an unpleasant sensation of the skin that provokes the urge to scratch or rub. It is one of the most common presenting complaints in dermatology and general practice, and its impact on quality of life can be profound — ranging from mild annoyance to severe sleep disruption, psychological distress, and clinical depression. Pruritus may be localized or generalized, acute or chronic, and critically, it may occur with visible skin changes or with entirely normal-appearing skin. The distinction between these two broad categories is the cornerstone of clinical evaluation, as pruritus with normal skin is a red flag for underlying systemic disease.

The pathophysiology of pruritus involves a complex interplay of neural pathways, inflammatory mediators, and central nervous system processing. When the skin encounters allergens or irritating substances, irritation and inflammation of epidermal nerve endings — specifically unmyelinated C fibers and type A delta nociceptive fibers — triggers the release of histamine and other pruritogenic mediators. This produces the sensation of itch, which leads to scratching. Scratching causes further skin damage and inflammation, which in turn releases more pruritogens, perpetuating the so-called **itch-scratch cycle**. Breaking this cycle is a central goal of management.

Complications of chronic pruritus include excoriations and skin injury, secondary infections (bacterial, viral, and fungal), scarring, and pigmentary changes. These complications can significantly compound the morbidity of the underlying condition.

Classification of Pruritus

Pruritus is classified into two major categories based on clinical examination:

- 1. Pruritus with skin changes** — where a primary dermatological condition is visible on the skin.
- 2. Pruritus with normal skin** — where no primary skin lesion is present, raising suspicion for systemic, neurological, or psychogenic causes.

This classification directs the entire investigative and management pathway.

Localized Pruritus with Skin Changes

When pruritus is accompanied by visible skin changes and is confined to specific areas, the differential diagnosis is broad and includes inflammatory dermatoses, infections, infestations, and environmental triggers.

Inflammatory Dermatoses

Several common inflammatory skin conditions present with localized pruritus and characteristic skin findings:

- **Eczema (atopic dermatitis):** A chronic, relapsing inflammatory condition characterized by pruritic, erythematous, scaly patches, often in flexural distribution. It is associated with a personal or family history of atopy (asthma, allergic rhinitis, eczema).
- **Psoriasis:** Well-demarcated, erythematous plaques with silvery scale, commonly affecting extensor surfaces (elbows, knees), scalp, and lumbosacral area. Pruritus is variable but can be significant.
- **Lichen planus:** Characterized by polygonal, purple, pruritic, flat-topped (planar) papules and plaques of unknown etiology. The surface may show fine white lines known as Wickham striae. Commonly affects the wrists, ankles, and oral mucosa.
- **Dermatitis herpetiformis:** An intensely pruritic, papulovesicular eruption classically located on the elbows, buttocks, and knees. It is associated with gluten-sensitive enteropathy (celiac disease) and may present with gastrointestinal symptoms such as bloating. Direct immunofluorescence shows granular IgA

deposition at the dermal papillae.

KEY TEACHING POINTS

Dermatitis herpetiformis is the cutaneous manifestation of celiac disease — all patients should be screened for gluten-sensitive enteropathy.

Lichen planus can involve the nails (longitudinal ridging, pterygium formation) and oral mucosa — always examine these sites.

Psoriasis is not always pruritic, but when it is, it significantly impacts quality of life and should not be dismissed.

Insect Bites, Infestations, and Environmental Causes

- **Insect bites/stings:** Can cause localized pruritic papules and wheals. Repeated scratching may lead to **nodular prurigo**, characterized by persistent itching with lichenified papules and nodules that are notoriously resistant to treatment.
- **Head lice (Pediculosis capitis):** Infestation causes intense scalp pruritus. Diagnosis is confirmed by finding live lice or viable nits on hair shafts.
- **Contact dermatitis:** Either irritant or allergic in etiology. Presents with erythema, vesicles, and pruritus localized to the site of contact with the offending agent. Common allergens include nickel, fragrances, and preservatives.
- **Polymorphic light eruption:** An acute allergic reaction to sunlight affecting sun-exposed skin. Presents with pruritic papules, plaques, or vesicles hours after UV exposure, typically in spring or early summer.

MCQ TRAPS & PITFALLS

Nodular prurigo is a **consequence** of chronic scratching, not a primary diagnosis — always look for an underlying cause.

Polymorphic light eruption spares the face and hands (chronically sun-exposed areas) — it affects areas with **intermittent** sun exposure.

Contact dermatitis can mimic eczema clinically — a detailed exposure history is essential.

Urticaria, Fungal Infections, and Anogenital Pruritus

- **Urticaria (hives) and angioedema:** Urticaria presents with transient, pruritic, erythematous wheals that migrate and resolve within 24 hours. Angioedema involves deeper dermal and subcutaneous swelling, particularly of the face and lips. Acute urticaria is commonly triggered by infections, drugs (especially NSAIDs and antibiotics), or food allergens.
- **Fungal infections:** Particularly **tinea pedis** (athlete's foot), which causes interdigital scaling, maceration, and intense pruritus. Other forms include tinea corporis (ringworm) and tinea cruris. Diagnosis is confirmed by potassium hydroxide (KOH) preparation and fungal culture.
- **Anogenital pruritus:** Pruritus ani and pruritus vulvae are common complaints. Causes include hemorrhoids, anal fissures, candidiasis, pinworms (*Enterobius vermicularis*), contact dermatitis, and

lichen sclerosus. In children, pinworm infestation is a leading cause of nocturnal perianal pruritus.

IMPORTANT

Urticarial wheals that persist >24 hours at the same site should raise suspicion for urticarial vasculitis — biopsy is warranted.

Pruritus ani with nocturnal worsening in a child is classic for pinworm (*Enterobius vermicularis*) — perform the tape test.

Tinea infections are frequently misdiagnosed as eczema — always perform KOH preparation before starting topical steroids.

Common Causes of Localized Pruritus with Skin Changes

Condition	Key Clinical Features	Diagnostic Clue
Eczema (atopic dermatitis)	Flexural erythema, scaling, lichenification	Personal/family history of atopy
Psoriasis	Well-demarcated plaques with silvery scale	Auspitz sign, nail pitting
Lichen planus	Purple, polygonal, flat-topped papules	Wickham striae, oral involvement
Dermatitis herpetiformis	Intensely pruritic papulovesicles on extensor surfaces	Granular IgA on immunofluorescence
Urticaria	Transient migratory wheals, pruritic	Individual lesions resolve <24 hours
Tinea pedis	Interdigital scaling, maceration, pruritus	Positive KOH preparation
Contact dermatitis	Erythema, vesicles at contact site	Exposure history, patch testing
Polymorphic light eruption	Pruritic papules on sun-exposed skin after UV exposure	Seasonal, spares chronically exposed areas

Generalized Pruritus with Skin Changes

Generalized pruritus accompanied by widespread skin changes suggests a more extensive dermatological process or a systemic condition with cutaneous manifestations.

Common Generalized Inflammatory and Infectious Causes

- **Atopic eczema (generalized):** Widespread erythematous, scaly, excoriated patches with lichenification in chronic cases. Often begins in childhood and may be associated with the atopic triad.
- **Scabies:** Caused by the mite *Sarcoptes scabiei*. Presents with intense generalized pruritus that is characteristically worse at night. Burrows, interdigital web spaces, wrists, axillae, waistline, and genitalia are commonly affected. Secondary eczematization and infection are common.
- **Pityriasis rosea:** A self-limiting condition beginning with a single "herald patch" followed by a generalized eruption of oval, scaly plaques along skin cleavage lines (Christmas tree pattern) on the

trunk. Mild pruritus is common. Resolves spontaneously in 6–8 weeks.

• **Drug eruptions:** Numerous medications can cause generalized pruritic rashes. Common culprits include antibiotics (penicillins, sulfonamides), NSAIDs, allopurinol, and anticonvulsants. Morphology varies from maculopapular exanthem to severe reactions like Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN).

KEY TEACHING POINTS

Scabies pruritus is worse at night because mite activity increases with warmer skin temperatures in bed.

Pityriasis rosea can mimic secondary syphilis — always perform serological testing (RPR/VDRL) to exclude this.

Drug eruptions should be considered in **any** patient presenting with acute generalized pruritus and rash — take a thorough medication history including over-the-counter and herbal products.

Cutaneous T-Cell Lymphoma (Mycosis Fungoides)

• **Mycosis fungoides:** The most common type of cutaneous T-cell lymphoma. It presents with chronic, progressive, pruritic skin lesions that evolve through stages: patch stage (flat, scaly, erythematous patches mimicking eczema or psoriasis) → plaque stage (raised, indurated plaques) → tumor stage (nodular masses that may ulcerate). Pruritus is often severe and refractory to standard treatments. Diagnosis is confirmed by skin biopsy showing epidermotropic atypical lymphocytes with Pautrier microabscesses.

IMPORTANT

Mycosis fungoides is frequently misdiagnosed as chronic eczema or psoriasis for years before the correct diagnosis is made.

Any patient with "eczema" that is refractory to standard therapy and progressively worsening should prompt consideration of cutaneous T-cell lymphoma.

Sézary syndrome is the leukemic variant, characterized by erythroderma, lymphadenopathy, and circulating atypical T cells (Sézary cells).

Generalized Pruritus with Skin Changes — Key Differentiating Features

Condition	Distribution	Key Feature	Duration
Scabies	Interdigital webs, wrists, axillae, waistline, genitalia	Burrows, nocturnal pruritus	Until treated
Pityriasis rosea	Trunk (Christmas tree pattern)	Herald patch precedes eruption	6–8 weeks (self-limiting)
Drug eruption	Generalized, often symmetric	Temporal relationship to new medication	Resolves with drug withdrawal
Cutaneous T-cell lymphoma	Patch → plaque → tumor progression	Refractory pruritus, progressive course	Chronic, progressive
Atopic eczema (generalized)	Flexures, face (infants), widespread	Atopic history, chronic relapsing	Chronic

Pruritus with Normal Skin — A Red Flag for Systemic Disease

When a patient presents with generalized pruritus and the skin appears entirely normal on examination, this is a critical clinical finding. The absence of primary skin lesions shifts the differential diagnosis away from primary dermatological conditions and toward systemic, neurological, and psychogenic causes. This scenario demands a thorough history, comprehensive physical examination, and targeted investigations.

Hematological and Lymphoproliferative Causes

- **Polycythemia vera (PV):** A myeloproliferative neoplasm characterized by increased red blood cell mass. Generalized pruritus is a hallmark symptom, often triggered by water contact (aquagenic pruritus). Patients may also present with plethora, headache, dizziness, and thrombotic events. Diagnosis is confirmed by elevated hemoglobin/hematocrit, JAK2 V617F mutation (present in >95% of cases), and bone marrow biopsy.
- **Hodgkin lymphoma:** Pruritus can be an early and sometimes the only presenting symptom of Hodgkin lymphoma, preceding the diagnosis by months. The pruritus is often generalized and severe. Associated features include painless lymphadenopathy (especially cervical), B symptoms (fever, night sweats, weight loss), and alcohol-induced nodal pain (pathognomonic but rare).
- **Iron deficiency anemia:** Even in the absence of frank anemia, iron deficiency can cause generalized pruritus. The mechanism is not fully understood but may relate to changes in epidermal innervation and mast cell function.

KEY TEACHING POINTS

Aquagenic pruritus (itching triggered by water contact) is highly suggestive of polycythemia vera — check a full blood count.

Generalized pruritus without rash in a young adult with B symptoms should prompt evaluation for Hodgkin lymphoma.

Iron deficiency can cause pruritus even before anemia develops — check ferritin levels.

Hepatobiliary and Cholestatic Causes

- **Cholestatic liver disease:** Intrahepatic or extrahepatic cholestasis causes generalized pruritus due to the accumulation of bile acids and other pruritogens in the skin. Conditions include primary biliary cholangitis (PBC), primary sclerosing cholangitis (PSC), obstructive jaundice (gallstones, pancreatic carcinoma, cholangiocarcinoma), and intrahepatic cholestasis of pregnancy.
- **Primary biliary cholangitis (PBC):** An autoimmune condition causing progressive destruction of intrahepatic bile ducts. Pruritus may precede jaundice by months to years. Associated with fatigue, xanthelasma, and other autoimmune conditions (Sjögren syndrome, thyroiditis). Diagnosis: elevated alkaline phosphatase, positive anti-mitochondrial antibodies (AMA) in >95% of cases.
- **Intrahepatic cholestasis of pregnancy:** Presents in the third trimester with generalized pruritus, typically starting on the palms and soles. Liver enzymes are elevated. Resolves after delivery but carries

a risk of fetal complications (preterm delivery, stillbirth).

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PBC predominantly affects middle-aged women (female:male ratio ~9:1) — a young man with cholestatic pruritus is more likely to have PSC.

Intrahepatic cholestasis of pregnancy resolves after delivery — reassure the patient but monitor liver function and fetal well-being.

Pruritus in obstructive jaundice is caused by bile salt deposition in the skin — it resolves with relief of obstruction.

Chronic Kidney Disease (Uremic Pruritus)

- **Uremic pruritus:** Affects up to 40–50% of patients with end-stage renal disease (ESRD) on dialysis. The pruritus is often generalized, worse at night, and can be debilitating. The exact pathogenesis is multifactorial, involving uremic toxin accumulation, immune dysregulation, opioid system imbalance, and peripheral neuropathy. It is associated with increased mortality and reduced quality of life.

IMPORTANT

Uremic pruritus may improve with optimization of dialysis adequacy, correction of calcium-phosphate imbalance, and management of secondary hyperparathyroidism.

Gabapentin and pregabalin have shown efficacy in uremic pruritus — consider these agents in refractory cases.

Emollients and phototherapy (UVB) are first-line non-pharmacological interventions.

Endocrine Causes

- **Hyperthyroidism:** Generalized pruritus is a recognized feature of thyrotoxicosis, likely due to increased skin blood flow and temperature. Other symptoms include weight loss, heat intolerance, tremor, palpitations, and anxiety.
- **Diabetes mellitus:** Pruritus in diabetes may be generalized or localized (especially anogenital, due to candidiasis). Mechanisms include autonomic neuropathy, dry skin (xerosis), and immune dysfunction.
- **Carcinoid syndrome:** Caused by serotonin-secreting neuroendocrine tumors. Presents with flushing, diarrhea, wheezing, and right-sided valvular heart disease. Pruritus may accompany the flushing episodes.

Systemic Causes of Pruritus with Normal Skin

System	Condition	Key Associated Features	Key Investigation
Hematological	Polycythemia vera	Aquagenic pruritus, plethora, thrombotic events	FBC, JAK2 mutation
Hematological	Hodgkin lymphoma	B symptoms, painless lymphadenopathy	Lymph node biopsy, CT/PET

System	Condition	Key Associated Features	Key Investigation
Hematological	Iron deficiency	Fatigue, koilonychia, pallor	Ferritin, iron studies
Hepatobiliary	PBC	Fatigue, xanthelasma, cholestatic LFTs	AMA, liver biopsy
Hepatobiliary	Obstructive jaundice	Dark urine, pale stools, jaundice	LFTs, ultrasound/MRCP
Renal	Uremic pruritus	ESRD on dialysis, disturbed sleep	Urea, creatinine, eGFR
Endocrine	Hyperthyroidism	Weight loss, tremor, heat intolerance	TSH, free T4
Endocrine	Diabetes mellitus	Polyuria, polydipsia, recurrent infections	Fasting glucose, HbA1c
Malignancy	Solid tumors (paraneoplastic)	Weight loss, anorexia, cachexia	Age-appropriate cancer screening

Neurological and Psychogenic Causes of Pruritus

Neuropathic Pruritus

Neuropathic pruritus arises from damage or dysfunction anywhere along the itch pathway — from peripheral nerves to the central nervous system. Unlike inflammatory pruritus, it is not mediated by histamine and therefore does not respond to antihistamines.

- **Notalgia paresthetica:** A localized pruritus of the interscapular region (T2–T6 dermatomes) caused by posterior rami nerve entrapment. Presents with persistent itching, burning, or tingling in a well-defined area of the upper back. May develop secondary hyperpigmentation from chronic scratching.
- **Brachioradial pruritus:** Localized pruritus of the lateral forearm, often associated with cervical spine pathology (cervical radiculopathy, degenerative disc disease). Thought to be caused by nerve root compression affecting C5–C6 dermatomes.
- **Postherpetic pruritus:** Can follow herpes zoster (shingles) in the affected dermatome. May be more troublesome than pain in some patients and is often refractory to treatment.

KEY TEACHING POINTS

Neuropathic pruritus does NOT respond to antihistamines — consider gabapentin, pregabalin, or topical capsaicin.

Brachioradial pruritus is associated with sun exposure in addition to cervical spine disease — patients may report worsening in summer.

Notalgia paresthetica is frequently misdiagnosed as tinea versicolor or post-inflammatory hyperpigmentation — consider the diagnosis when the itch precedes the visible skin change.

Psychogenic Pruritus

- **Psychogenic (functional) pruritus:** A diagnosis of exclusion. Pruritus is attributed to psychological factors such as anxiety, depression, obsessive-compulsive disorder, or delusional disorders (e.g., delusional parasitosis). The itch is often described as crawling, biting, or stinging. Patients may bring "specimens" (skin debris, fibers) in a matchbox — the classic "matchbox sign" of delusional parasitosis.
- **Delusional parasitosis (Ekbom syndrome):** A fixed, false belief of skin infestation with parasites. Patients have no insight into the psychiatric nature of their condition. Management requires a delicate, non-confrontational approach. Low-dose antipsychotics (e.g., risperidone 0.5–2mg daily, olanzapine 2.5–5mg daily) may be effective.

IMPORTANT

Psychogenic pruritus is a diagnosis of exclusion — systemic causes must be thoroughly investigated and excluded first.

Delusional parasitosis patients will see multiple dermatologists and refuse psychiatric referral — building rapport is essential.

The "matchbox sign" (patient bringing collected skin debris as evidence of infestation) is pathognomonic for delusional parasitosis.

Diagnostic Approach to Pruritus

A systematic approach to pruritus evaluation is essential to avoid missing serious underlying conditions.

History Taking

A thorough history should include:

- **Onset and duration:** Acute (<6 weeks) vs. chronic (≥6 weeks) pruritus
- **Distribution:** Localized vs. generalized
- **Temporal pattern:** Nocturnal worsening (scabies, uremic pruritus), seasonal variation (polymorphic light eruption)
- **Triggers:** Water contact (polycythemia vera), heat, stress, exercise
- **Associated symptoms:** Fever, weight loss, night sweats (lymphoma), jaundice (cholestasis), polyuria/polydipsia (diabetes)
- **Medication history:** Prescription, over-the-counter, herbal, and recreational drugs
- **Occupational and travel history:** Exposure to irritants, travel to endemic areas (scabies, parasitic infections)
- **Past medical history:** Atopy, thyroid disease, liver disease, renal disease, malignancy
- **Family history:** Atopy, hematological malignancies
- **Psychosocial assessment:** Stress, anxiety, depression, substance use

Physical Examination

- **Full skin examination:** Under good lighting, including interdigital web spaces, scalp, nails, oral mucosa, and anogenital area
- **Characterize any lesions:** Primary lesions (macules, papules, vesicles, wheals, nodules) vs. secondary changes (excoriations, lichenification, scarring)
- **Lymph node examination:** Cervical, axillary, inguinal — lymphadenopathy may indicate lymphoma
- **Hepatosplenomegaly:** May indicate hematological malignancy or liver disease
- **Thyroid examination:** Goiter, nodules

Investigations

The investigation panel should be guided by clinical suspicion:

OSCE PEARLS

Always examine the skin under good lighting — subtle findings (burrows in scabies, Wickham striae in lichen planus) are easily missed.

In pruritus with normal skin, the examination is your most powerful tool — a thorough systemic examination can reveal hepatosplenomegaly, lymphadenopathy, or thyroid enlargement.

Ask about water contact triggering itch — this single question can lead to the diagnosis of polycythemia vera.

Recommended Investigations for Pruritus with Normal Skin

Investigation	Rationale	Conditions Screened
Full blood count (FBC)	Anemia, polycythemia, eosinophilia	Iron deficiency, PV, parasitic infection, lymphoma
Liver function tests (LFTs)	Cholestatic pattern (elevated ALP, GGT)	PBC, PSC, obstructive jaundice
Renal function (U&E, eGFR)	Uremia	CKD/ESRD
Thyroid function (TSH, fT4)	Hyperthyroidism	Graves disease, toxic nodular goiter
Fasting glucose/HbA1c	Diabetes mellitus	Type 1 and type 2 diabetes
Ferritin and iron studies	Iron deficiency	Iron deficiency anemia
Serum protein electrophoresis	Paraproteins	Myeloma, lymphoma
Anti-mitochondrial antibodies (AMA)	Autoimmune cholestasis	Primary biliary cholangitis
Chest radiograph	Mediastinal lymphadenopathy	Hodgkin lymphoma, sarcoidosis
Urinalysis	Hematuria, proteinuria	Renal disease, myeloma
HIV serology	Immunosuppression-related pruritus	HIV infection
Stool ova and parasites	Parasitic infestation	Strongyloides, hookworm

Management of Pruritus

Management of pruritus is directed at the underlying cause when identified, combined with symptomatic relief measures.

General Measures and Emollients

- **Emollients:** The cornerstone of symptomatic management. They restore the skin barrier, reduce transepidermal water loss, and provide a protective layer. Apply liberally and frequently, especially after bathing. Ointment-based emollients are more effective than creams or lotions.
- **Bathing advice:** Use lukewarm (not hot) water, limit bath/shower time, use soap-free cleansers, and apply emollients immediately after patting dry.
- **Clothing:** Wear loose-fitting, cotton clothing. Avoid wool and synthetic fabrics that can irritate the skin.
- **Environmental measures:** Maintain cool, humidified indoor environment. Use a humidifier in dry climates or during winter months.

Topical Therapies

- **Topical corticosteroids:** Effective for inflammatory causes of pruritus (eczema, psoriasis, contact dermatitis). Use the lowest potency sufficient for the affected area. Avoid prolonged use on thin skin (face, flexures, genitalia).
- **Topical calcineurin inhibitors:** Tacrolimus 0.03–0.1% ointment and pimecrolimus 1% cream. Useful for facial and flexural eczema, and as steroid-sparing agents.
- **Menthol/camphor preparations:** Provide a cooling sensation that can temporarily relieve pruritus. Available in various concentrations (0.5–5% menthol).
- **Topical capsaicin:** Depletes substance P from sensory nerve endings. Useful for localized neuropathic pruritus (notalgia paresthetica, brachioradial pruritus). Initial burning sensation is common and limits compliance.

Systemic Therapies

- **Antihistamines:** First-line for histamine-mediated pruritus (urticaria, allergic reactions). Non-sedating antihistamines (cetirizine 10mg daily, loratadine 10mg daily, fexofenadine 180mg daily) are preferred for daytime use. Sedating antihistamines (hydroxyzine 25mg at night, chlorpheniramine 4mg TID) may be useful for nocturnal pruritus.
- **Gabapentin/Pregabalin:** Effective for neuropathic pruritus and uremic pruritus. Gabapentin starting dose 100–300mg at night, titrated up to 900–1800mg/day in divided doses. Pregabalin 75mg BD, titrated to 150–300mg BD.
- **SSRIs:** Paroxetine (20mg daily) and sertraline (50–100mg daily) have shown benefit in psychogenic pruritus and paraneoplastic pruritus.
- **Phototherapy (UVB):** Narrowband UVB is effective for uremic pruritus, pruritus of Hodgkin lymphoma, and other refractory causes. Typically administered 2–3 times per week.

IMPORTANT

Cholestyramine is the first-line treatment for cholestatic pruritus — it binds bile acids in the gut. Administer 2 hours before or after other medications to avoid binding.

Rifampicin is second-line for cholestatic pruritus — monitor LFTs closely as it is hepatotoxic.

Naltrexone (opioid antagonist) is effective in cholestatic and uremic pruritus — start at 12.5–25mg daily and titrate up to 50mg BD.

Pharmacological Management of Pruritus by Etiology

Etiology	First-Line Treatment	Second-Line Treatment
Urticaria	Non-sedating antihistamines (up to 4x standard dose)	Omalizumab, cyclosporine
Eczema/dermatitis	Topical corticosteroids + emollients	Topical calcineurin inhibitors, phototherapy
Scabies	Permethrin 5% cream (whole body, repeat in 7 days)	Ivermectin 200µg/kg PO (single dose, repeat in 7 days)
Uremic pruritus	Emollients, optimize dialysis, gabapentin	Phototherapy (UVB), nalfurafine
Neuropathic pruritus	Gabapentin or pregabalin	Topical capsaicin, amitriptyline
Psychogenic pruritus	SSRIs (paroxetine, sertraline)	Low-dose antipsychotics (risperidone)
Cholestatic pruritus	Cholestyramine 4–16g/day	Rifampicin 150–600mg BD, naltrexone
Polycythemia vera	Aspirin, phlebotomy, hydroxyurea	Ruxolitinib (JAK inhibitor)

Special Populations

Pruritus in Pregnancy

Pruritus in pregnancy has a unique differential diagnosis that must be considered:

- **Intrahepatic cholestasis of pregnancy (ICP):** Third trimester, pruritus starting on palms and soles, elevated bile acids and LFTs. Resolves after delivery. Risk of fetal complications.
- **Polymorphic eruption of pregnancy (PUPPP):** Most common dermatosis of pregnancy. Presents in the third trimester with pruritic urticarial plaques starting in striae and spreading to the trunk. Resolves after delivery. No fetal risk.
- **Pemphigoid gestationis:** Autoimmune blistering disorder of pregnancy. Presents with intensely pruritic urticarial plaques progressing to tense vesicles and bullae, typically starting periumbilically. Associated with HLA-DR3/DR4. May flare with subsequent pregnancies and with oral contraceptive use.

KEY TEACHING POINTS

Intrahepatic cholestasis of pregnancy is the only pregnancy-specific dermatosis that carries fetal risk — early delivery may be indicated.

Pemphigoid gestationis can be confused with PUPPP initially — the presence of vesicles/bullae and periumbilical onset should raise suspicion.

PUPPP is more common in first pregnancies and multiple gestations — it does not recur in subsequent pregnancies.

Pruritus in the Elderly

- **Xerosis (dry skin):** The most common cause of pruritus in the elderly. Age-related reduction in sebaceous gland activity and skin barrier function leads to dry, flaky, pruritic skin. Management focuses on regular emollient use and avoidance of irritants.
- **Medication-induced pruritus:** Elderly patients are often on multiple medications, many of which can cause pruritus (opioids, ACE inhibitors, statins, diuretics).
- **Herpes zoster:** More common in the elderly and immunocompromised. Pruritus may precede the vesicular eruption (prodromal phase) or persist as postherpetic pruritus.

Practice Questions

Q: A 55-year-old man presents with a 3-month history of generalized pruritus that is worse after bathing. He reports headaches and dizziness. On examination, his skin appears normal. His face appears plethoric. Full blood count shows hemoglobin 192 g/L, hematocrit 0.58, and white cell count $14.2 \times 10^9/L$. JAK2 V617F mutation is positive. What is the most likely diagnosis?

- A. Chronic myeloid leukemia
- B. Polycythemia vera
- C. Essential thrombocythemia
- D. Iron deficiency anemia
- E. Secondary polycythemia

Answer: B

Explanation: The combination of aquagenic pruritus (worsening after bathing), plethora, elevated hemoglobin and hematocrit, and positive JAK2 V617F mutation is diagnostic of polycythemia vera. JAK2 V617F is present in >95% of PV cases. Aquagenic pruritus is a hallmark feature of PV and is present in up to 40% of patients.

Q: A 32-year-old woman presents with generalized pruritus and fatigue. She has noticed yellow patches around her eyelids. Liver function tests show elevated alkaline phosphatase (ALP 480 U/L) and gamma-glutamyl transferase (GGT 320 U/L), with normal bilirubin. Anti-mitochondrial antibodies are positive. What is the most likely diagnosis?

- A. Primary sclerosing cholangitis
- B. Hepatitis B infection

- C. Primary biliary cholangitis
- D. Drug-induced cholestasis
- E. Cholangiocarcinoma

Answer: C

Explanation: The presentation of a middle-aged woman with pruritus, fatigue, xanthelasma (yellow patches around eyelids), cholestatic liver enzyme pattern (elevated ALP and GGT), and positive anti-mitochondrial antibodies is classic for primary biliary cholangitis. AMA is positive in >95% of PBC cases. Pruritus may precede jaundice by months to years.

Q: A 68-year-old woman on hemodialysis for end-stage renal disease reports severe generalized pruritus that disrupts her sleep. Her skin examination is normal except for mild xerosis. Her dialysis adequacy (Kt/V) is 1.0 (target >1.2). Serum calcium is 2.15 mmol/L, phosphate is 2.0 mmol/L, and PTH is 45 pmol/L. What is the most appropriate next step in management?

- A. Start oral cetirizine 10mg daily
- B. Optimize dialysis adequacy and start gabapentin
- C. Refer for renal transplantation
- D. Start topical betamethasone valerate 0.1% cream
- E. Prescribe oral prednisolone 20mg daily

Answer: B

Explanation: Uremic pruritus is multifactorial. The first step is to optimize dialysis adequacy (current Kt/V is suboptimal at 1.0). Gabapentin is effective for uremic pruritus and is the most appropriate pharmacological intervention. Antihistamines are generally ineffective for uremic pruritus as it is not histamine-mediated. Topical steroids are not indicated for pruritus with normal skin.

Q: A 24-year-old man presents with a 2-week history of generalized pruritus and a rash. He initially noticed a single large, scaly, erythematous patch on his trunk, followed days later by multiple smaller oval lesions distributed along skin cleavage lines in a "Christmas tree" pattern. The lesions are mildly pruritic. What is the most likely diagnosis?

- A. Secondary syphilis
- B. Pityriasis rosea
- C. Guttate psoriasis
- D. Tinea corporis
- E. Pityriasis versicolor

Answer: B

Explanation: The classic presentation of pityriasis rosea includes: (1) a herald patch (single large scaly lesion) preceding the generalized eruption; (2) oval, scaly plaques distributed along skin cleavage lines in a Christmas tree pattern on the trunk; (3) mild pruritus; (4) self-limiting course (6–8 weeks). However, secondary syphilis can mimic this presentation — RPR/VDRL should always be performed to exclude it.

Q: A 45-year-old woman presents with persistent itching of the upper inner back (interscapular region) for 6 months. Examination shows a well-defined area of hyperpigmentation but no primary skin lesion. She has a history of cervical spondylosis. What is the most likely diagnosis?

- A. Tinea versicolor
- B. Notalgia paresthetica
- C. Contact dermatitis
- D. Lichen simplex chronicus
- E. Brachioradial pruritus

Answer: B

Explanation: Notalgia paresthetica is a localized neuropathic pruritus affecting the interscapular region (T2–T6 dermatomes), caused by posterior rami nerve entrapment, often associated with cervical spine pathology. The hyperpigmentation is secondary to chronic scratching. Brachioradial pruritus affects the lateral forearm (C5–C6), not the interscapular region.

Q: A 28-year-old woman in her 34th week of pregnancy presents with severe generalized pruritus that started on her palms and soles. She has no rash. Liver function tests show elevated bile acids (25 $\mu\text{mol/L}$) and mildly elevated transaminases (ALT 68 U/L, AST 55 U/L). What is the most important next step?

- A. Reassure and advise emollients only
- B. Start oral antihistamines and schedule delivery at 40 weeks
- C. Monitor bile acids and fetal well-being, consider early delivery
- D. Perform a skin biopsy
- E. Start oral prednisolone

Answer: C

Explanation: Intrahepatic cholestasis of pregnancy (ICP) presents in the third trimester with pruritus starting on palms and soles, elevated bile acids, and abnormal LFTs. ICP carries a risk of fetal complications including preterm delivery, meconium staining, and stillbirth. Management involves monitoring bile acids and fetal well-being, with consideration of early delivery (typically at 37–38 weeks or earlier if bile acids are significantly elevated). Ursodeoxycholic acid is the first-line treatment for symptom relief.